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Prepared Data Analysis & Insights

Executive Summary

- This project analyzes a sample Sales dataset (50,000 records) covering transactions, customers, and regions. Using SQL, Python, and visualization libraries, we performed data wrangling, exploratory analysis, predictive modeling, and interactive dashboard creation. Results highlight regional sales patterns and a classification model predicting high-value customers with 86% accuracy.

Introduction

- The project aims to understand sales behavior, identify key drivers of revenue, and predict customer value. Business motivation: enhance decision-making, optimize marketing strategies, and improve forecasting accuracy.

Data Collection & Wrangling

- • Dataset: 50,000 sales transactions (CSV format)
- • Key fields: Customer_ID, Region, Product, Sales, Date
- • Cleaning steps:
 - - Removed duplicates (2%)
 - - Handled missing values (imputed 5% with median)
 - - Converted dates into year, month, and quarter

Exploratory Data Analysis (EDA)

- • Univariate analysis: Sales distribution is right-skewed
- • Bivariate analysis: Higher sales in North & West regions
- • Correlations: Strong positive correlation between discount and sales volume
- • Outlier detection: Identified extreme values in product pricing

Predictive Analysis Methodology

- • Task: Predict if a customer is 'High Value' (top 20% revenue)
- • Models tested: Logistic Regression, Random Forest, XGBoost
- • Evaluation metrics: Accuracy, Precision, Recall, F1-score
- • Final Model: Random Forest (max depth=10, n_estimators=200)
- • Rationale: Balanced accuracy and interpretability

EDA Visualization Results

- • Regional Sales Trends: North contributed 40% of total revenue
- • Product Analysis: Top 5 products accounted for 60% sales
- • Seasonal Trends: Q4 peak in sales across all regions
- • Customer Segmentation: High-value customers tend to buy electronics

EDA SQL Results

- Example Query: Identify top 5 regions by sales
- `SELECT Region, SUM(Sales) AS Total_Sales`
- `FROM transactions`
- `GROUP BY Region`
- `ORDER BY Total_Sales DESC;`
- Insight: North region leads with 40% contribution, followed by West (30%).

Interactive Map (Folium)

- • Created a Folium map plotting sales density across regions
- • North region markers show higher concentration
- • Interactive tooltips display total sales per region
- • Helped visualize geographic disparities in revenue

Plotly Dash Dashboard

- • Built an interactive dashboard with filters for product, region, and time
- • Visuals: Bar charts, line charts, KPIs
- • Features: Drill-down into regional sales trends
- • Benefit: Real-time decision support for managers

Predictive Analysis Results

- • Random Forest Classifier performance:
 - - Accuracy: 86%
 - - Precision: 0.83
 - - Recall: 0.79
 - - F1-score: 0.81
- • Confusion matrix showed balanced performance across classes
- • ROC-AUC score: 0.89

Conclusion

- • Sales concentrated in North & West regions
- • Seasonal spikes in Q4 sales
- • High-value customers linked to electronics purchases
- • Predictive model achieved 86% accuracy in identifying high-value customers
- • Insights can guide marketing and resource allocation

Creativity & Insights

- • Used interactive Folium maps and Dash dashboards for advanced visualization
- • Applied predictive analytics for customer segmentation
- • Combined SQL + Python workflows for flexible analysis
- • Deliverables go beyond standard reporting, offering actionable insights